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ASK

1 How to control Maxon Motor by using ROS_CANopen

epos2

epos_hardware

Hi all,

asked Dec 16 '16

CJ.Nilson 13 ●3 ●4 ●8

updated Dec 27 '16

I am a newbie to ROS, I would like to use ROS_CANopen package to connect with my robot, I want to set different modes of operation for position and current control.

After that, I would like to use MoveIt to control my robot. However, I checked from previous [ROSAnswer](#) about ROS_CANopen can not find more detail tutorials on how use it.

Here is my hardware configuration:

1. A [USB-to-CAN compact](#) made by IXXAT is used to communicate between my laptop and the robot. The Linux driver called IXXAT [socketcan driver](#).
2. The robot uses Maxon EPOS2 with CANopen. eds file is generated from EPOS studio.

OS: Ubuntu 14.04 with indigo installed. I already tested with socketcan driver by using can-utils i.e. cansend can0 xxx. Therefore, I can just use CANopen protocol-301 and 402 with command-line to control Maxon motor, in the meantime, use sockketcn_dump to verify command and status.

How to use Ros_CANopen? could someone give me a hint to do next steps or some examples I could refer to? or any package to run roslaunch to control ROS_CANopen?

Many thanks,

Comments

- 1 Excuse me, I now have a USB-to-CAN device (kvaser leaf light v2), a motor! I want to know how to use ROS_CANopen and socketcan_driver to control motor rotation? Can you tell me what to do? This problem has been bothering me for a long time, please!

 [FAN](#) (Aug 16 '19)

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1 Answer

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It should be more or less straightforward:

answered Dec 16 '16

Mathias Lüdtké  1526 ●2 ●25 ●26

updated Dec 16 '16

1. Setup SocketCAN (you already have this running):
http://wiki.ros.org/socketcan_interfa...
2. Prepare URDF with sane limits (this is required for now, will be optional in the future):
<http://wiki.ros.org/urdf/Tutorials/Cr...>
3. Prepare EDS/DCF, take special care for the PDO mapping (http://wiki.ros.org/canopen_402), use transmission type 1 (every sync)
4. Prepare driver config

- prepare chain config (http://wiki.ros.org/canopen_chain_nod...)
 - configure motor-specific settings, especially unit conversions (http://wiki.ros.org/canopen_motor_node)
5. Prepare controller config
 - list of controllers (documentation varies, http://wiki.ros.org/ros_controllers)
 - add a `required_drive_mode` to each motion controller, it takes the number of the mode to be used (http://wiki.ros.org/canopen_402#Drive...)
 6. Prepare a launchfile for `canopen_motor_node`
 - upload driver config as private parameters
 - upload controller config into namespace
 - upload robot description
 7. rosservice call `[/namespace]/driver/init` (http://wiki.ros.org/canopen_chain_nod...)
 8. start/switch the controllers (`rqt_controller_manager`, `ontroller_manager`, `spawner` etc.)

The EDS file parser is not really bullet proof (and EDS file format is rather flexibel) and might fail with some error indication. In addition `ros_canopen` enforces strict typing, some EDS files list non-standard types for some objects (e.g. signed instead of unsigned). This must be fixed as well in the EDS.

A basic driver config looks like:

```
bus:
  device: can0
sync:
  interval_ms: 10 # 100Hz
  overflow: 0
defaults:
  eds_pkg: my_package
  eds_file: "config/MyDescription.dcf" # the parser reads EDS and DCF
nodes:
  my_joint:
    id: 1
```

controller config example: https://github.com/ipa320/canopen_tes...

Fairly complex examples: https://github.com/ipa320/schunk_robots/

It would be great if you could turn this into a tutorial on ROS Wiki based on your experiences :)

link

Comments

Thank you for the replies. I still have some questions regarding your replies.

- item 2: I have created URDF to describe elements fo my robot, but why need URDF in `ros_canopen`
- item 5: Why `ros_canopen` package need `ros_control`? Could you briefly introduce the main idea for using `ros_control`

 CJ.Nilson (Dec 20 '16)

I'm trying to refer "`schunk_robots` and `canopen_test_utils`" as a model for item 4, 5 and 6. Which packages do you recommend for my case? Thanks you very much.

 CJ.Nilson (Dec 20 '16)

The URDF is used for getting the joint limits and to verify the names of the joints.

More information on `ros_control` can be found in the wiki: http://wiki.ros.org/ros_control And `schunk_robots` contains full-fledged examples.. If in doubt have a look at both.

 Mathias Lüttke (Dec 20 '16)

When I tried to do item 7 you listed, I met the [failure](#) during initializing the lwa4d, have you ever meet this before ? Thanks for your reply

 [CJ.Nilson](#) ([Dec 23 '16](#))

- 1 Hello CJ and Mathias, please, could you help me with a similar problem in this question: [#q292882](#)
Many thanks, Jorge

 [jdeleon](#) ([Jun 1 '18](#))

[@Mathias Lüdtke](#) , When you say use transmission type 1, that is for both RxPDO and TxPDO ?

 [Cyril_J](#) ([Dec 6 '18](#))

Yes, for both. This is how we run it with Care-O-bot. It is not mandatory, but it enforces a deterministic data flow. Otherwise the timing would be arbitrary and might result in (small) jumps.

 [Mathias Lüdtke](#) ([Dec 10 '18](#))

hi [@Mathias Lüdtke](#) , following your steps to successfully control the motor, is user supposed to write and implement some additional hardware interface for Maxon motor, or it is yet included in ros_canopen packages?

 [martinlucan](#) ([Jul 6 '21](#))

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